3.1 The Chain Rule

Name _____ Period ____ Date _____

AP Calculus 3.1 · Student sheet

I can…

- identify the outer and inner functions of a composite function.
- use the Chain Rule, including the Power Rule for functions.
- use the Chain Rule with e^u, trig functions, and tables of values.

Vocabulary · fill in the missing word

Composite function — $(f \circ g)(x) = f(g(x))$: the _____ of g is the input of f

Chain Rule — $\frac{d}{dx} f(g(x)) =$ _____

Power Rule for functions — $\frac{d}{dx} [u(x)]^n =$ _____

Part 1 · Expand or not?

$f(x) = (2x + 1)^2$

Method	f′(x)	Matches?
Expand: $4x^2 + 4x + 1$		
Guess: $2(2x + 1)$		
Fixed guess		

Part 2 · Notes and examples (fill in each note, then work the example)

Chain Rule: if $y = f(u)$ and $u = g(x)$, then $dy/dx = (dy/du) \cdot$ _____.

Example 1

$y = u^5$ and $u = x^2 + 1$. Find $\frac{dy}{dx}$.

Chain Rule (function form): $d/dx\ f(g(x)) = f'(g(x)) \cdot$ _____.

Example 2

$f(2) = 5$, $f'(2) = 3$, $g(1) = 2$, $g'(1) = -4$, $f'(5) = 6$. $h(x) = f(g(x))$. Find $h'(1)$.

Power Rule for functions: $d/dx\ [u(x)]^n = n[u(x)]^{n-1} \cdot$ _____.

Example 3

Find $f'(x)$ for $f(x) = (3x^2 - 5)^4$, then find $f'(1)$.

Exponential and trig: $\frac{d}{dx} e^u = e^u \cdot u'$, $\frac{d}{dx} \sin u =$ _____ $\cdot u'$.

Example 4

Find each derivative.

a. $y = e^{x^2 - 3x}$

b. $y = \sin(4x)$

Example 5

Find the tangent line to $y = (x^2 - 3)^3$ at $x = 2$.

Part 3 · Partner practice

Partner 1

$f(x) = \cos(3x)$. Find $f'\left(\frac{\pi}{6}\right)$.

Partner 2

(AP® Practice) If $f(x) = (x^2 - 3)^4$, then $f'(2) =$

(A) 4 (B) 16 (C) 32 (D) 64